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BE (8th Semester)

Examination, April-May, 2012

Branch : Mechanical Engineering

ROBOTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Question (a) and (b) are compulsory which carries 2 and 4 marks respectively. Answer any one question (c) or (d) which carries 10 marks. Draw neat sketch diagram wherever necessary.

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Q. 1. (a) What do you mean by major and miner axes of a robotic manipulator ? 2

(b) How do you specify a robot ? 4

(c) What are the basic components of robotic system ? Explain the function of each of the components with neat sketch. 10

(d) Explain with neat sketch spherical-coordinate of the robot. Where within the work Envelop is the tool tip precision the highest ? 10

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Q. 2. (a) How many DOFs (degree of freedoms) are required to position an end effector at any point in 3-D space ? 2

(b) What do you mean by homogeneous composite transformation ? 4

(c) If F and M be two initially coincident fixed and mobile orthonormal coordinate frames. Find the first mobile coordinate about fixed coordinate $[M^1]^f$ after composite transformation. If mobile coordinate M

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translate along fixed second coordinate F^2 and then rotate about fixed third coordinate F^3 by angle π radians. Sketch the equivalent axis which maps M coordinate into F. 10

(d) Find composite matrix for Yaw-Pitch-Roll transformation. 10

Q. 3. (a) What are the arm parameters used to define link coordinate ? 2

(b) What are the basic steps used to find arm equation ? 4

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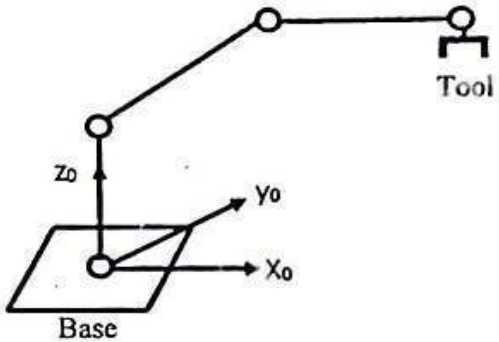
(c) Find out the arm equation by D-H algorithms of Alpha II robotic arm as shown in figure. 10



(d) Draw link coordinate diagram and write down the kinematic parameters in the form of a table for four-axis SCARA robot as shown in figure. 10

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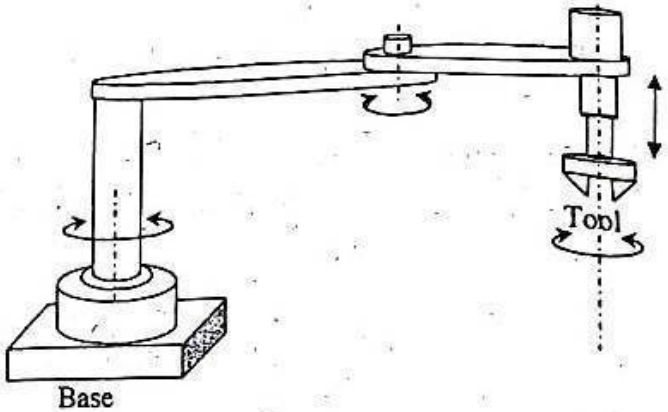

Q. 4. (a) What do you mean by sensing devices ? 2

(b) What are the different types of sensors ? Classify them. 4

(c) What do you mean by template matching ? Write an algorithm for template matching. 10

(d) What do you mean by edge detection ? Write an algorithm for edge detection. 10

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Q. 5. (a) Which robot is suitable for pick and place ? 2

(b) How can you justify the robot is better for industry ? Enumerate any four points. 4

(c) What are the non industrial applications of the robot ? Explain at least five non industrial applications of robot. 10

(d) State characteristics of work which promote application of robots. Discuss robot application for assembly and inspection. 10

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